

Numerical Analysis homework IV

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I. Convert 477 to a normalized FPN

Easy to see

$$477 = (111011101)_2$$

so

$$477 = (1.11011101) \times 2^8$$

II. Convert $\frac{3}{5}$ to a normalized FPN

Easy to see

$$\frac{3}{5} = (0.10011001\dots)_2$$

so

$$\frac{3}{5} = (1.0011001\dots)_2 \times 2^{-1}$$

III. Prove an equation

We know $x = 1.0 \times \beta^e$, and $e \in (L, U) \Rightarrow e+1, e-1 \in [L, U]$. Then

$$x_R = 1.00\dots 01 \times \beta^e \quad x_L = (\beta-1)(\beta-1)(\beta-1)\dots(\beta-1)(\beta-1) \times \beta^{e-1}$$

Both have p number. Then

$$x - x_L = \beta^{e-1-p+1} \Rightarrow RHS = \beta^{e-p+1} = x_R - x$$

IV. Two normalized FPNs adjacent to x

We know in IEEE 754 single-precision protocol, $x = (1.0011001\dots)_2 \times 2^{-1}$. Then

$$x_L = (1.0011001\dots 10011001)_2 \times 2^{-1} \quad x_R = (1.0011001\dots 10011010)_2 \times 2^{-1}$$

$\Rightarrow$

$$x - x_L = \frac{3}{5} \times 2^{-24} \quad x_R - x = x_R - x_L + x_L - x = 1 \times 2^{-23} - \frac{3}{5} \times 2^{-24} = \frac{2}{5} \times 2^{-24}$$

So

$$fl(x) = x_R \quad \delta = \frac{|fl(x) - x|}{|x|} = \frac{1}{3} \times 2^{-23}$$

V. New method of rounding off number

By the definition in the course, the unit roundoff represents the maximul round off error around 1, where the floating number is the densest. By the new method of rounding off number : the badest case is the number around 1 is $1+2^{-23}$ in IEEE-754. So

$$\epsilon_u^{new} = 2^{-23} = 2\epsilon_u$$

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VI. Estimate the precision lost in the subtraction

We know:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + O(x^6)$$

$\Rightarrow$

$$1 - \cos \frac{1}{4} = 2^{-5} + \epsilon \quad (\epsilon \ll 2^{-5})$$

So the subtraction will lose 5 bits of precision.

VII. Symmetric Polynomials

Method 1:

$$1 - \cos x = 2 \sin^2\left(\frac{x}{2}\right)$$

Then we know near 0 $2 \sin^2(\frac{x}{2})$ is small. So without the subtraction of two closed number. It will NOT cause a catastrophic cancellation.

Method 2:

$$1 - \cos x = \frac{1 - \cos^2 x}{1 + \cos x} = \frac{\sin^2 x}{1 + \cos x}$$

For the denominator, both of them are the number closed to 1, which will NOT cause a catastrophic cancellation. Meanwhile for the numerator, it's just a square of small number, which also avoid a subtraction of two closed number.

VIII. Condition numbers of functions

We know $\text{cond}_f(x) = \left| \frac{f'(x)x}{f(x)} \right|$

VIII.a

When $\alpha = 0$ the function is trivial. We only consider the case $\alpha \neq 0$

$$\text{cond}_f(x) = \left| \frac{\alpha(x-1)^{\alpha-1}x}{(x-1)^\alpha} \right| = \left| \frac{\alpha x}{x-1} \right|$$

So the condition number is large around $x = 1$.

VIII.b

$$\text{cond}_f(x) = \left| \frac{\frac{1}{x}x}{\ln x} \right| = \left| \frac{1}{\ln x} \right|$$

So the condition number is large around $x = 1$.

VIII.c

$$\text{cond}_f(x) = \left| \frac{e^x x}{e^x} \right| = |x|$$

So the condition number is large when $x \rightarrow \pm\infty$.

VIII.d

$$\text{cond}_f(x) = \left| \frac{x \frac{-1}{\sqrt{1-x^2}}}{\arccos x} \right| = \left| \frac{x}{\arccos x \sqrt{1-x^2}} \right|$$

So the condition number is large around $x = \pm 1$.

IX. Function $f(x) = 1 - e^{-x}$

IX.a

$$\text{cond}_f(x) = \left| \frac{f'(x)x}{f(x)} \right| = \left| \frac{xe^{-x}}{1 - e^{-x}} \right| = \left| \frac{x}{e^x - 1} \right|$$

When $x \in (0, 1]$, $\text{cond}_f(x) = \frac{x}{e^x - 1}$, besides we know $e^x \geq 1 + x$. Meanwhile, $\lim_{x \rightarrow 0} \frac{x}{e^x - 1} = 1$. So $\text{cond}_f(x) \leq 1$, when $x \in [0, 1]$.

IX.b

We know $f_A(x) = (1 - (1 + \delta_1)e^{-x})(1 + \delta_2)$, and $\delta_1, \delta_2 \leq \epsilon_u$, so

$$|\delta(x)| = \left| \frac{f_A(x) - f(x)}{f(x)} \right| = \left| \frac{\delta_2 - (\delta_1 + \delta_2 + \delta_1\delta_2)e^{-x}}{1 - e^{-x}} \right| \leq \left| \frac{\delta_2 + (\delta_1 + \delta_2 + \delta_1\delta_2)e^{-x}}{1 - e^{-x}} \right| \approx \left| \frac{\epsilon_u e^x + 2\epsilon_u}{e^x - 1} \right| = \epsilon_u \left| \frac{e^x + 2}{e^x - 1} \right|$$

Therefore:

$$\text{cond}_A(x) \leq \frac{|\delta(x)|}{\epsilon_u \text{cond}_f(x)} = \frac{|e^x + 2|}{|x|}$$

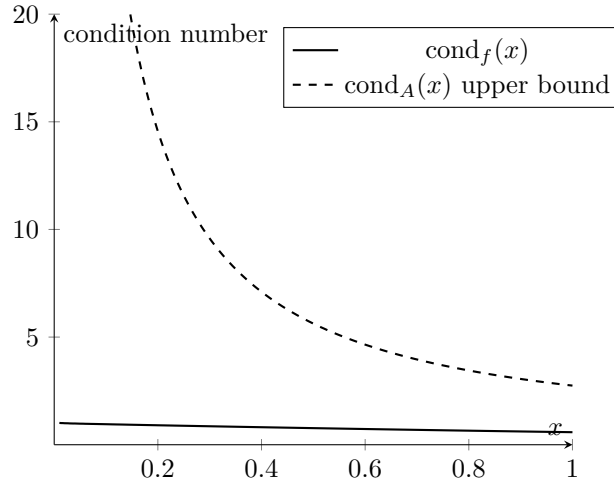


Figure 1: Condition numbers on $[0, 1]$

when $x \rightarrow 0$, the condition number $\text{cond}_A(x) \rightarrow \infty$, it's ill-posed. In the other points, the algorithm works.

X. Prove a lemma

We know

$$\|A\|_2 = \max_{\|x\|=1} \|Ax\|_2 = \max_{\|x\|=1} \sqrt{x^* A^* A x}$$

Attention $A^* A$ is symmetric matrix, which can be diagonalized. Let $\{v_i\}_{1 \leq i \leq n}$ be the eigen vectors of $\{\mu_i\}_{1 \leq i \leq n}$ of $A^* A$ and o.n.b by the way (eigen vectors in different eigen vector space are orthogonal). Then every x can be written as $\sum_{i=1}^n \alpha_i v_i$, so

$$\|A\|_2 = \max_{\|x\|=1} \sqrt{x^* A^* A x} = \max_{\sum_{i=1}^n \alpha_i^2 = 1} \sqrt{\sum_{i=1}^n \mu_i^2 \alpha_i^2} \leq \sigma_{\max}$$

Meanwhile

$$\|A\|_2 \geq \max_{1 \leq i \leq n} \sqrt{v_i^* A^* A v_i} = \sigma_{\max}$$

Then $\|A\|_2 = \sigma_{\max}$. And notice that $(A^{-1})^* A^{-1} = (A A^*)^{-1}$. Besides by linear algebra $|\lambda E - AB| = |\lambda E - BA|$. So we know AA^* has the same eigen value as $A^* A$. Then $\|A^{-1}\|_2 = \frac{1}{\sigma_{\min}}$, since the eigen value of $(A A^*)^{-1}$ is $\{\frac{1}{\mu_i}\}_{1 \leq i \leq n}$.

So $\text{cond}_2 A = \frac{\sigma_{\max}}{\sigma_{\min}}$.

If $A = A^*$, then $\exists$ unitary matrix U , s.t.

$$UAU^* = \text{diag}(\lambda_1, \dots, \lambda_n) \Rightarrow UA^*AU^* = UAU^*UAU^* = \text{diag}(\lambda_1^2, \dots, \lambda_n^2)$$

Then $\text{cond}_2 A = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$.

If $A^*A = E \Rightarrow \mu_i = 1, \forall i$, so $\sigma_{\max} = \sigma_{\min} = 1$. Then $\text{cond}_2 A = 1$.

XI. Find root of a polynomial

We know

$$q(r) = 0 \Rightarrow r^n + \sum_{i=0}^{n-1} a_i r^i = 0 \Rightarrow \frac{\partial r}{\partial a_i} q'(r) + r^i = 0$$

$$\text{cond}_f(\mathbf{x}) = \|A(\mathbf{x})\|_1 = \|(a_{ij}(\mathbf{x}))\|_1 = \max_{0 \leq i \leq n-1} \left| \frac{a_i \frac{\partial r}{\partial a_i}}{r} \right| = \max_{0 \leq i \leq n-1} \left| \frac{a_i r^{i-1}}{q'(r)} \right|$$

Besides for root r

$$W(x) := \sum_{i=1}^m (x - i) \Rightarrow \text{cond}_f(\mathbf{x}) = \max_{0 \leq i \leq m-1} \left| \frac{r^{i-1} a_i}{W'(r)} \right| = \max_{0 \leq i \leq m-1} \left| \frac{r^{i-1} a_i}{\prod_{j=1, j \neq r}^m (r - j)} \right|$$

Specially $r = m = 20$, $\text{cond}_f(\mathbf{x}) \approx 2.0 \times 10^{10}$.

$r = m = 30$, $\text{cond}_f(\mathbf{x}) \approx 2.0 \times 10^{16}$.

$r = m = 40$, $\text{cond}_f(\mathbf{x}) \approx 1.5 \times 10^{22}$.

Then by comparison, i see that if a question's condition is large enough, it's almost impossible to find a numerical algorithm that make the condition small.

XII. Given a counter example

Choose $\beta = 10, p = 2$. Then given $1/(2/7)$.

$$fl_p(1/fl_p(2/7)) = fl_p(1/0.29) = 3.4$$

$$fl_{2p}(1/fl_{2p}(2/7)) = fl_{2p}(1/0.2857) = 3.500 \rightarrow 3.5$$

Then get a contradiction.

XIII. Accuracy of computing root in IEEE 754

The answer is NO.

In IEEE 754 $\beta = 2, p = 24, L = -126, U = 127$. We know at least after $\lceil \frac{6}{\lg 2} \rceil + 1 = 20$ steps, we acquire the 10^{-6} accuracy. But the number $\in [128, 129]$, expressed as $(1.a_1 \dots a_{23})_2 \times 2^7$. The accuracy of the number is at most $2^{7-23-1} = 2^{-17}$, which at most ensure the accuracy of 17 steps. So we can't compute the root with absolute accuracy $< 10^{-6}$.

XIV. Explain a phenomenon

We know, for a pair every close adjacent points, may as well $x_1 = x_0 + h$, in this interval:

$$s(x) = a(x - x_0)^3 + b(x - x_0)^2 + c(x - x_0) + d$$

So we have

$$S[a, b, c, d]^T = M$$

For the curve is continuous, we know there are two line elements of S are

$$(0, 0, 0, 1) \quad (h^3, h^2, h, 1)$$

which is almost linear dependent when $h \rightarrow 0$, so the $\sigma_{\min} \rightarrow \infty$. Then the condition of S , $\text{cond}_2(S) \rightarrow \infty$, which will make the problem ill. Therefore will get an inaccurate result.